CVARC Logger

User Manual

Version 1.44

Program by W6CSM

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1. Introduction

CVARC Logger is a Windows desktop application for logging amateur radio contacts (QSOs). It is built to handle your everyday, average contacts just as smoothly as specialized activity like contest exchanges, SOTA summit activations, and POTA park activations. The design goal is fast, live, in-the-chair logging: enter a callsign, look it up online, and save the contact in a keystroke or two, while the log grid, awards tracking, and radio integration all update in the background without any extra effort on your part.

At a glance, CVARC Logger offers:

- Fast QSO entry with automatic online callsign lookup (Callook.info, QRZ.com, and QRZCQ.com).
- A Log Mode picker (Normal, Contest, SOTA, POTA) that reshapes the entry form to show only the fields the current activity actually needs.
- Live radio control (CAT) over USB via Hamlib's rigctld, or over the network to an Elecraft K4, auto-filling Frequency, Band, Mode, and TX Power as you tune.
- Contest logging support: a running Sequence number for the exchange serial, plus ARRL Sweepstakes and Field Day fields (Precedence, Check, Class) and SKCC member numbers.
- Multiple station profiles, so different callsigns, grid squares, or operating locations each keep their own settings.
- A searchable, sortable, and fully customizable QSO log grid with multi-select, bulk delete, and copy/paste to and from a spreadsheet.
- ADIF import and export, compatible with QRZ Logbook, LoTW, and virtually every other logging program.
- Award progress tracking for DXCC, Worked All States (WAS), Mountain Goat (SOTA), and Parks on the Air (POTA), computed automatically from your log.
- Automatic QSO capture from WSJT-X (relayed through GridTracker2 over UDP), and real-time QSO broadcast out to GridTracker2 for live mapping.
- Support for multiple independent log files, so you can keep separate logs for different callsigns, contests, or operating locations.

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2. Getting Started

2.1 Launching CVARC Logger

Run CvarcLogger.exe. The title bar shows the running version, for example “CVARC Logger v1.44”. On first launch, a new log database (cvarclogger.db) is created automatically in the same folder as the .exe, so a portable install—run from a USB drive or a synced folder—carries its data with it.

A station profile is required before the main window will open. If no profile exists yet (including on a brand-new log), the Station Profiles window (Chapter 3) opens first, and the program cannot be used until at least one profile with a callsign and a valid UTC Offset has been saved. The first profile you save becomes the default.

If you are upgrading from an older install that already kept its database under %LOCALAPPDATA%\CVARC Logger, that existing database continues to be used automatically, rather than starting a second, empty one next to the .exe.

2.2 The Main Window

The main window is divided into four areas, top to bottom:

- Banner — the club logo and the title “Conejo Valley Amateur Radio Club Logger” across the top.
- Menu bar — the File, Stations, View, and Tools menus, covered throughout this manual.
- QSO Entry form — where you fill in and log a new contact.
- QSO Log grid — every contact logged so far, searchable and sortable.
- Status bar — the total QSO count, the current log file name, and a “Program by W6CSM” credit.

2.3 Quick Start

1. Open Stations > Manage Station Profiles and create a profile for your callsign (see Chapter 3).
2. Type a callsign into the QSO Entry form and click Lookup to fill in the contact's details.
3. Fill in Band, Mode, and signal reports as needed.
4. Click Log QSO (or press Enter) to save the contact. The form clears and the new QSO appears at the top of the log grid below.

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3. Station Profiles

A station profile represents one operating identity—typically your callsign, plus the location and settings that go with it. Every QSO you log is tied to a station profile, and that profile's details (callsign, grid, county, and so on) are copied onto the QSO at the moment you log it. Later changes to a profile therefore never rewrite the history of contacts you have already logged.

Open Stations > Manage Station Profiles to create, edit, or delete profiles.

3.1 Profile Fields

- Callsign (required) — the station's callsign.
- Operator Callsign — needed only when the operator differs from the station callsign; when set, it takes priority over the OP name as the more specific identifier.
- My Grid Square, My State, My County, and QTH — your operating location.
- OP (Operator Name) — a free-text fallback used only when Operator Callsign is blank.
- SKCC # — your Straight Key Century Club member number, shown automatically as “My SKCC #” on the entry form so you don't have to re-enter your own number for each contact.
- UTC Offset (required; hours, e.g. -5 or +5.5) and Observes Daylight Saving Time — used together to compute each QSO's Local Time for display.
- Default station profile — the profile pre-selected in the QSO Entry form the next time the program starts.

Required fields are marked with an asterisk in the window, and a “* Required” note appears at the bottom. The default profile is shown in bold in the profile list on the left.

3.2 Managing Profiles

Select a profile from the list on the left to edit it, or click New Profile to start a blank one. Click Save to keep your changes, Delete to remove the selected profile, or Close when you are finished.

Station profiles carry over automatically when you create a new log with File > New Log (Chapter 9), so switching logs never means re-entering your callsign and grid square.

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4. Logging a QSO

4.1 The QSO Entry Form

The entry form at the top of the main window is where you build and save each contact. Depending on the selected Log Mode (Section 4.2) and your column choices (Section 5.2), it can show any of the following fields:

- Log Mode — Normal, Contest, SOTA, or POTA (Section 4.2).
- Station — which station profile this QSO is logged under.
- Callsign — the contact's callsign, automatically shown in upper case.
- Date/Time (UTC), Local Time, and Time Off (UTC) — the Date/Time and Local Time fields tick live once per second while idle; type over them to enter a manual or backdated time.
- Band, Freq (MHz), Mode, and Sub-Mode — Band and Mode are editable dropdowns, so you can type a value that isn't on the list. Sub-Mode appears only for modes that have one (SSB, and the digital DATA modes).
- RST/S and RST/R — signal reports sent and received.
- Name, Grid, City, State, County, and Country — the contact's details, mostly filled by Lookup.
- ARRL Section, CQ Zone, and ITU Zone.
- My SOTA / SOTA and My POTA / POTA — summit and park references for activations (Chapter 7).
- OP, QTH, and TX Power (W) — OP and QTH default from the selected station profile but can be overridden for a single contact.
- Comment — a single free-text note (Section 8.3).
- Contest and SKCC fields — Sequence #, Precedence, Check, Class, SKCC #, and My SKCC # (Section 4.5).

Callsign, Station, Date/Time (UTC), Local Time, Band, Mode, and Sub-Mode are always available. Every other field is optional and is shown or hidden using the same Choose Columns checkboxes as the log grid (Section 5.2): unchecking a column there also removes its input box from the entry form, so the form only ever shows the fields you actually track.

The field your cursor is in is highlighted with a bold blue border, so it is always obvious which box you are typing in across the wide form.

The available Bands are 160m, 80m, 60m, 40m, 30m, 20m, 17m, 15m, 12m, 10m, 6m, 2m, 1.2M, and 70cm. The available Modes are SSB, CW, FM, AM, FT8, FT4, RTTY, PSK31, DMR, D-STAR, and DATA.

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4.2 Log Modes

The Log Mode picker at the top-left of the entry form reshapes the form to match what you are doing, hiding fields that don't apply so there is less to skip past on every contact:

- Normal — the full form, with every field you have enabled. Use this for everyday operating.
- Contest — adds the contest exchange fields (Sequence #, Precedence, Check, and Class) alongside State, for ARRL Sweepstakes and Field Day style logging (Section 4.5).
- SOTA — moves the summit fields up near the top of the form and hides the fields an activation doesn't need (Time Off, City, County, Country, Comment, and SKCC).
- POTA — the same idea for park activations, with the park fields moved up and the unused fields hidden.

Switching modes only changes which fields are visible; nothing already typed is lost, and the always-on fields stay put in every mode.

4.3 Callsign Lookup

Click Lookup—or simply log the QSO (Section 4.4)—to query the configured online lookup services for the callsign you entered. A successful lookup fills in Name, Grid, Country, State, County, and City, but only where those fields are currently empty, so it never overwrites something you have already typed by hand.

ARRL Section is derived automatically from State and County rather than looked up directly. CQ and ITU Zone are resolved from the contact's grid square when one is available (accurate even for large, multi-zone countries like the USA); if no grid square is known, the zone falls back to the nominal zone for the contact's DXCC entity.

Which services are queried, and the QRZ / QRZCQ credentials two of them require, are configured under Tools > Lookup (Section 10.1).

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4.4 Logging the QSO

Click Log QSO, or press Enter while focus is anywhere on the entry form, to save the contact. If you have not already looked up the current callsign (or you changed the callsign after looking it up), CVARC Logger runs the lookup automatically first—you never need to remember to click Lookup yourself.

If a dropdown (Band, Mode, Sub-Mode, or Station) is open when you press Enter, the first Enter closes the dropdown and a second Enter actually logs the QSO. This is normal Windows combo-box behavior, not a fault in the program.

After logging, the form clears and is ready for the next contact, and the new QSO appears immediately at the top of the log grid. A number of fields deliberately keep their value for the next contact instead of clearing—these are the “static” fields, tinted pale yellow and labelled “static” in small grey text: Station, Band, Mode, Sub-Mode, RST sent and received, My SOTA, My POTA, OP, QTH, TX Power, and My SKCC #. Frequency also carries over. Each carried-over field can be edited normally when a contact needs something different; a tooltip on each one explains the behavior.

4.5 Contest and SKCC Logging

Switch the Log Mode to Contest to log an ARRL Sweepstakes, Field Day, or similar exchange. The following fields appear:

- Sequence # — the running contact serial number sent in the exchange. Click Start to begin the count at 1; it then increments automatically after every logged QSO. Click Reset to zero it and stop counting until you press Start again. The value is saved as the QSO's sent serial number (ADIF STX) and can be shown in the log grid as the “Seq #” column.
- Precedence — a dropdown listing the ARRL precedence categories, each shown with its full definition so you don't have to memorize them.
- Check — the two-digit check (hover the field label for an explanation of what the Check is).
- Class — the Field Day operating class.

SKCC (Straight Key Century Club) numbers are supported independently of contest mode. Your own SKCC number is set once in your station profile (Section 3.1) and is shown automatically as a read-only “My SKCC #” field; the contacted station's number is entered per QSO in the SKCC # field. All of these fields—Sequence, Precedence, Check, Class, and both SKCC numbers—round-trip through ADIF on import and export.

4.6 Radio Control (CAT)

If CAT is enabled and configured (Tools > CAT Control, Section 10.2), click Connect the CAT to link the entry form to your radio. The CAT icon beside the button changes to show when a connection is live. Once connected:

- Frequency, Band, and Mode update automatically as you tune the radio (polled roughly every 1.5 seconds).
- For SSB, the USB/LSB sub-mode is filled in automatically from the radio.
- For digital modes the radio can only report a generic “DATA” state, so you still choose the specific sub-mode (FT8, FT4, RTTY, and so on) yourself.
- TX Power (W) is filled in from the radio's transmit-power reading, scaled by the Max Power (W) value you set for that radio (Section 10.2).

Check Pause auto-fill to temporarily stop CAT from overwriting Frequency, Band, and Mode—useful while you are mid-way through typing details for the next contact and don't want the fields to jump. Click Disconnect the CAT to end the session. A status message beside the button reports the connection state and any errors; if the radio connection is lost, polling stops automatically and the status message explains why.

4.7 Clearing the Log

The Clear Database button, at the right of the entry form, deletes every QSO from the current log. It is deliberately styled dark red—not the green Log QSO button—and asks a Yes/No confirmation first, so it can't become an accidental one-click erase. There is no undo, so use it only when you really do want to empty the current log; to preserve contacts first, export them to ADIF (Section 8.2) or switch to a different log (Chapter 9).

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5. The QSO Log Grid

Every logged QSO appears as a row in the grid at the bottom of the main window, most recent contacts first, with a chronological row number on the left (oldest contact is number 1).

5.1 Searching

Type into the Search box above the grid to filter the log as you type. Click Refresh to reload the grid from the log file.

5.2 Choosing and Reordering Columns

Click Columns to choose which fields are shown. Callsign and Station Callsign are always shown and cannot be removed; every other column is optional, and All / None buttons let you toggle them in one click. The picker lists columns alphabetically by name: Date/Time (UTC), Local Time, Time Off (UTC), Band, Mode, Sub-Mode, Freq, Freq Rx, RST S/R, Name, Grid, City, State, County, Country, Continent, ARRL Section, CQ Zone, ITU Zone, TX Power, QSL S/R, LoTW S/R, QSL Via, Operator, My Grid, My State, My County, QTH, OP, My SOTA, SOTA, My POTA, POTA, Comment, and the contest/SKCC columns Precedence, Check, Class, SKCC #, My SKCC #, and Seq # (all hidden by default).

Drag a column header left or right to reorder it. Both your column choices and your column order are remembered automatically and restored the next time you open CVARC Logger. Because unchecking an optional column also hides that field's box on the entry form (Section 4.1), the Columns picker effectively tailors the whole program to the fields you care about.

5.3 Editing a QSO

Double-click a row, or select it and click Edit, to open the Edit QSO window (Chapter 6).

5.4 Selecting Multiple QSOs

- Ctrl + right-click a row to toggle that row into or out of the current selection.
- Shift + right-click a row to select every row between the last-touched row and the one you clicked.

A normal left-click selects a single row as usual. The selected row stays clearly highlighted in blue even while you are typing in the entry form.

5.5 Deleting QSOs

Select one or more rows (Section 5.4) and click Delete. Every selected QSO is deleted at once. This cannot be undone, so double-check the selection first.

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5.6 Copy and Paste

- Ctrl + C copies the selected rows as tab-separated text, using only the columns currently visible, in their current left-to-right order.
- Ctrl + V reads tab-separated rows from the clipboard and logs each one as a new QSO, matching cells back to columns using that same visible column layout.

This makes it easy to round-trip a block of QSOs through Excel or another spreadsheet, or to copy contacts between two CVARC Logger windows—as long as the column layout hasn't changed between the copy and the paste. The Local Time column is read-only (it is calculated from the UTC time and the station's UTC offset), so it is simply skipped on paste.

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6. Editing a QSO

The Edit QSO window—opened from the log grid (Section 5.3)—exposes every field of a logged contact for correction, including fields that aren't shown as grid columns. The window is resizable and scrolls if needed. Pressing Return saves, just as it does on the main entry form.

6.1 Contact Details

- Date/Time On (UTC) and Date/Time Off (UTC) — typed in yyyy-MM-dd HH:mm format; the field enforces this format as you type.
- Callsign, Band, Mode, Sub-Mode, Freq (MHz), and Freq Rx (MHz).
- RST Sent, RST Rcvd, Name, Grid, City, State, County, and Country.
- ARRL Section, Continent, CQ Zone, ITU Zone, and TX Power (W).
- Station Callsign, Operator, My Grid, My State, My County, QTH, and OP.
- UTC Offset (h) and Observes DST.
- Comment — a single free-text field (Section 8.3 explains how this maps on ADIF import/export).

A Lookup button is available in this window too, in case you need to re-run the online lookup while correcting a contact.

6.2 QSL Status

A separate QSL Status section tracks paper and electronic confirmation:

- QSL Sent / QSL Sent Date and QSL Rcvd / QSL Rcvd Date, plus QSL Via (the callsign a card is routed through, if any).
- LoTW Sent / LoTW Sent Date and LoTW Rcvd / LoTW Rcvd Date.

Date fields here use yyyy-MM-dd format. Click Save to keep your changes or Cancel to discard them.

7. Awards Progress

Open View > Awards Progress to see your progress toward several common awards, all computed automatically from your logged QSOs.

7.1 DXCC

Lists every DXCC entity, whether you have worked it, and whether it is confirmed, along with running Worked and Confirmed totals.

7.2 WAS (Worked All States)

Lists all 50 US states with Worked and Confirmed checkmarks, and running totals out of 50.

7.3 Mountain Goat (SOTA)

Tracks progress toward SOTA's 1,000-point Mountain Goat activator award. Add a summit code (for example W6/CT-003) and click Add to start tracking it; its point value and official summit name are looked up automatically from the SOTA summit list. Any summit logged via the entry form's My SOTA field is picked up automatically as well, the next time the Awards Progress window is opened.

A summit counts as activated—its points included in the running total—only once at least 4 contacts have been logged from it on the same UTC date. Until then, its Points column shows the value it would be worth in parentheses, for example “(4)”, and the Activation Date/Time column always shows the date and time of the first contact, so partial progress toward the 4-contact minimum stays visible even before the summit qualifies.

Click Refresh Summit List to force an immediate re-download of the official SOTA summit list instead of waiting for its automatic 30-day refresh. Select a row and click Delete Selected to stop tracking a summit.

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7.4 Parks on the Air (POTA)

Tracks progress toward POTA's activator award tiers, based on the total number of unique parks activated: Bronze (10), Silver (20), Gold (30), Platinum (40), Diamond (50), Ruby (100), and Emerald (125). Add a park reference (for example US-0001) and click Add to start tracking it; its name is looked up automatically. Any park logged via the entry form's My POTA field is picked up automatically as well, the next time the Awards Progress window is opened.

A park counts as activated once at least 10 different callsigns have been contacted from it on the same UTC date—repeat contacts with the same station don't count toward the 10. A separate Kilo award tracks 1,000 cumulative QSOs logged at one single park (repeat activations of that park all count toward it, unlike the tiers above); the Kilo column checks off automatically once a tracked park's Total QSOs reaches 1,000.

Click Refresh Park Info to re-check live park names from POTA's own API for every park currently being tracked. Unlike SOTA, POTA has no single bulk export to refresh a full offline list from, so this only re-checks parks already on your list, not the full worldwide park directory. Select a row and click Delete Selected to stop tracking a park.

7.5 About DXCC Accuracy

DXCC entity resolution uses a bundled, community-assembled approximation of the ARRL DXCC prefix list, not the official ARRL data. If a QSO shows the wrong entity, correct it directly on that QSO in the log (Chapter 6); the Awards totals update automatically from whatever is in the log.

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8. Importing and Exporting Logs (ADIF)

ADIF (.adi / .adif) is the standard interchange format for amateur radio logs, supported by QRZ Logbook, LoTW, and virtually every other logging program.

8.1 Importing

Choose File > Import ADIF, then pick a .adi or .adif file. Every record with a callsign is imported as a new QSO, and a summary such as “Imported 42 of 42 record(s)” is shown when the import finishes.

- DXCC entity is always re-resolved from the callsign during import rather than trusted from the source file, so an entity code the source software used that CVARC Logger doesn't recognize can never abort the import partway through.
- When importing a QRZ Logbook export, QSL confirmation status is read from QRZ's own confirmation fields rather than the standard QSL_RCVD tag, since QRZ's exports commonly leave that tag blank even for confirmed contacts.

8.2 Exporting

Choose File > Export ADIF, pick a destination file, and every QSO in the current log is written out. The suggested filename defaults to the current log's name plus a date/time stamp (for example FieldDay2026_20260722_153045.adi), so exports from different logs are self-identifying and repeated exports don't overwrite each other. A summary such as “Exported 128 QSO(s)” is shown when the export finishes.

8.3 The Comment Field

CVARC Logger has a single free-text “Comment” field per QSO, which round-trips through ADIF's COMMENT tag on both import and export—matching QRZ and most other logging software, which also use COMMENT rather than NOTES for this.

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9. Managing Multiple Logs

A “log” in CVARC Logger is a single database file (.db) containing QSOs and station profiles. You can keep more than one—for example, separate logs for different callsigns, contests, or operating locations—and switch between them from the File menu.

9.1 Creating a New Log

Choose File > New Log, pick a file name and location, and CVARC Logger creates a fresh, empty log there. Your existing station profiles are copied into the new log automatically, so you don't need to re-enter them. CVARC Logger then asks to restart to switch over to the new log.

9.2 Opening an Existing Log

Choose File > Open Log, pick an existing .db file, and CVARC Logger verifies that it is a genuine CVARC Logger database before switching to it. As with creating a new log, a restart is required to complete the switch.

9.3 Where Logs Are Stored

By default, a brand-new log is created next to CvarcLogger.exe, so a portable copy of the program (for example on a USB drive) carries its data with it. The current log's file name is always shown in the status bar at the bottom of the main window; hover over it to see the full path.

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10. Settings

CVARC Logger's settings live in two windows on the Tools menu: Lookup (callsign lookup and the GridTracker2 / WSJT-X log relays) and CAT Control (radio control). Both windows resize with their content, and credentials entered in either are stored encrypted (Windows DPAPI) on this machine only—never written into the QSO log itself.

10.1 Callsign Lookup (Tools > Lookup)

The lookup services are checked in a fixed order: QRZ.com first, then QRZCQ.com, then Callook.info. Each service is skipped automatically if it isn't configured, or if the results already gathered have everything it could add—most notably County, which neither Callook nor QRZCQ provides at all. You don't choose a single “preferred” service; whichever services are configured are tried in that order until the contact's details are complete.

QRZ.com XML Data API

Requires a paid QRZ XML Data subscription. Enter your QRZ username and password and click Save Credentials; use Show password to reveal a saved password, or Clear to remove the stored credentials. A status line reports whether credentials are saved.

QRZCQ.com XML API

Requires a Premium QRZCQ account. Enter your QRZCQ username and password and click Save Credentials, the same way as for QRZ. Note that QRZCQ does not provide County, so that field still has to come from whichever other configured service supplies it.

10.2 Radio Control (Tools > CAT Control)

The CAT Control window has a CAT Source selector at the top with three choices—Off, USB (Hamlib), and Internet—that decides how the entry form's Connect the CAT button reads the radio. Only one source is active at a time. A single Save Settings button saves the whole window, and Test Connection checks whichever source is currently selected.

USB / Serial (Hamlib)

CVARC Logger talks to a USB- or serial-connected radio through Hamlib's rigctld, a small TCP server that Hamlib provides. It uses the copy of rigctld.exe bundled in the hamlib folder next to CvarcLogger.exe automatically—no separate Hamlib install is needed. To point it at a different copy instead (for example one already installed with WSJT-X or from hamlib.org), set the rigctld path to that copy's full location.

- Active radio — which of the configured radio slots below is currently in use.
- Refresh Ports — COM ports are re-scanned each time the window opens; use this button to re-scan without closing and reopening.

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- Each radio slot has its own Radio (Hamlib) selector, Hamlib Model ID, COM Port, Baud Rate, and Max Power (W). If your radio isn't in the Radio dropdown, run rigctl --list from a command prompt to find its Model ID and enter that directly.
- rigctld path — assumed to be on the system PATH unless you give a full path to rigctld.exe.
- rigctld TCP port — the port CVARC Logger connects to (default 4532).
- Launch rigctld automatically — starts rigctld for you when you click Connect the CAT, so you don't have to start it yourself beforehand.

The bundled default radio slots are:

Radio	Hamlib Model ID	Default COM Port	Default Baud Rate
Elecraft K4D	2047	COM3	38400
Yaesu FT-991A	1035	COM4	38400
Kenwood TS-890	2041	COM5	115200
Yaesu FT-920	1014	COM7	4800

About Max Power (W): Hamlib reports live transmit power as a 0.0–1.0 fraction of the radio's power ceiling, not real watts. Multiplying that fraction by Max Power (W) is what produces the TX Power (W) value that auto-fills on the entry form while CAT is connected (Section 4.6). Max Power is the one piece of that calculation CVARC Logger can't get from Hamlib or the radio, so you have to set it yourself to match your radio's actual rated (or user-configured) maximum output; it defaults to 100 and is never auto-detected or changed on its own. To calibrate, key up at a known power level, compare the radio's own meter against the TX Power (W) that CVARC Logger displays, and adjust Max Power up or down until the two agree.

Internet Control (Elecraft K4)

The Internet source reads frequency and mode from an Elecraft K4 over the network using the radio's native TCP command protocol (not Hamlib). Point Host / IP at the radio's address; to reach it over the internet, use your public IP or DDNS name with the port forwarded to the radio. A password is optional—the K4 needs none itself, so set one only if you reach the radio through an authenticated tunnel or gateway. A status line reports the connection state.

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10.3 GridTracker2

When enabled (in the Tools > Lookup window), CVARC Logger broadcasts every logged or edited QSO as a single ADIF record over UDP, so GridTracker2 can plot it on its map in real time. Point Host and Port at whatever GridTracker2's "Receive ADIF UDP (Broadcasts for Logging)" setting (its General tab) is listening on—127.0.0.1 and port 2240 by default. On the GridTracker2 side, set that port to 2240 and check its enabled box; this is the counterpart setting that actually receives what CVARC Logger sends.

Check Enable outbound log Broadcast to GridTracker2 to turn this on, set Host and Port as needed, and click Save GridTracker Settings. Use Send Test QSO to confirm GridTracker2 is receiving packets before you start operating.

10.4 WSJT-X

CVARC Logger can receive WSJT-X's logged-QSO data as relayed by GridTracker2, over UDP port 2238, and add those contacts to the log automatically. WSJT-X broadcasts to GridTracker2 first (on its own default port, 2237); GridTracker2 logs the contact and then forwards the same message on to CVARC Logger on port 2238. CVARC Logger does not listen to WSJT-X's broadcast directly.

Check Enable WSJT-X log receiving (in the Tools > Lookup window) to turn this on. QSOs picked up this way are deliberately not forwarded back to GridTracker2, which already has them from WSJT-X directly. All three programs—CVARC Logger, WSJT-X, and GridTracker2—must be running on the same PC. On the GridTracker2 side, under its General tab, enable "Forward UDP messages" and set its port to 2238; without this, CVARC Logger will never receive anything on that port even though WSJT-X and GridTracker2 are otherwise working normally.

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11. Keyboard Shortcuts & Quick Reference

Action	Shortcut / Control
Log the current QSO	Enter, anywhere in the QSO Entry form
Look up the current callsign	Click Lookup
Toggle a row into/out of multi-selection	Ctrl + Right-click a row in the log grid
Select a range of rows	Shift + Right-click a row in the log grid
Copy selected rows	Ctrl + C, with focus in the log grid
Paste rows as new QSOs	Ctrl + V, with focus in the log grid
Open the Edit QSO window	Double-click a row, or select it and click Edit
Delete selected QSOs	Select rows, then click Delete

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12. Tips & Troubleshooting

The Enter key seems to do nothing

If a Band, Mode, Sub-Mode, or Station dropdown is open, the first Enter press just closes it. Press Enter again to log the QSO.

CAT stopped updating Frequency / Band / Mode

Check the status message next to the Connect / Disconnect the CAT button—if the radio connection was lost, CVARC Logger stops polling automatically and explains why. Reconnect with Connect the CAT once the radio is reachable again. Also check whether Pause auto-fill is checked.

A contact shows the wrong DXCC entity or Country

The bundled DXCC prefix table is a community approximation, not official ARRL data (Section 7.5). Correct the Country field directly on the QSO via Edit (Chapter 6); the Awards totals update automatically.

New Log / Open Log doesn't seem to do anything immediately

Both actions require restarting CVARC Logger to actually switch the active log—you'll be prompted to restart right away, or you can restart later yourself.

County isn't filled in after Lookup

Callook and QRZCQ don't provide County at all; only QRZ.com's XML Data API does. If QRZ isn't configured, County has to be typed in manually.

TX Power (W) doesn't match the radio's own meter

Adjust the Max Power (W) field for that radio in Tools > CAT Control (Section 10.2); see that section for how to calibrate it.

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Appendix A: Version History

Selected highlights from the CVARC Logger changelog. Versions before 1.17 predate the changelog and are not itemized here.

Version	Highlights
1.46	Fixed a crash seeding the DXCC entity table on a brand-new database.
1.44	Added a Log Mode preset picker (Normal/Contest/SOTA/POTA) to the QSO Entry form, and a Contest-mode Sequence # field with Start/Reset buttons for the contest exchange serial number.
1.43	Added ARRL Sweepstakes/Field Day contest logging (Precedence, Check, Class) and SKCC number support, with a Precedence dropdown showing full ARRL definitions and a Check field tooltip. QSO Entry fields now center their values (except Callsign, Station Callsign, Operator Callsign, Freq, Freq Rx, and Comment). Log grid column picker and ADIF import/export gained the new contest/SKCC fields.
1.42	Fixed a bug where editing a QSO could change its recorded date/time. Log grid Name and Comment columns are now left-aligned. The Return key now saves in the Edit QSO window.
1.41	Frequency carries over between QSOs; a Clear Database button; clearer cursor/selection highlighting; ADIF exports named after the active log.
1.40	Internet Control (CAT) for a networked Elecraft K4. One CAT Source selector (Off/USB/Internet) with a single Save and Test button. Installer now offers Update vs Fresh.
1.39	Settings split into separate Lookup and CAT Control windows. Enlarged, fixed-row entry form with a tinted background; Columns picker gained All/None buttons.
1.38	Log grid: manual column resize, centered headers/data with divider lines and narrower columns. Time Off (UTC) field, centered field titles, uniform box heights. Show Password reveals a saved password; Station Profiles preloads the default; app data defaults to the

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	exe folder.
1.37	Clarified the Max Power (W)/rigctld manual wording; Station Profile required-field markers; Settings credential status text and Show password toggle; -none- radio slot option; new-install checkbox defaults.
1.36	Documented the Max Power (W) calibration field; removed hard page breaks that left large blank gaps between sections.
1.35	WSJT-X-logged QSOs now get an automatic callsign lookup; log grid row numbers count chronologically (oldest = 1); CAT now auto-fills TX Power (with a new per-radio Max Power calibration setting); CAT icon changes when connected.
1.34	WSJT-X log receiving now works via GridTracker2's UDP relay (port 2238) instead of competing with GridTracker2 for WSJT-X's own broadcast port. GridTracker2's outbound broadcast port changed to 2240 to match. Settings checkbox renamed for clarity.
1.33	Mountain Goat (SOTA) points shown as (#) until activated, plus Summit Name column. New Parks on the Air (POTA) award tab with Bronze-Emerald tiers and Kilo award. Refresh buttons for both SOTA and POTA data. QSO Entry form gained OP, QTH, and TX Power fields.
1.32	Requires a station profile on first launch; Mountain Goat (SOTA) award tracking tab with 4-contact activation rule; WSJT-X UDP log receiver; new club logo/icon; Choose Columns and log grid styling/sorting improvements.
1.31	QSO entry form's Date/Time (UTC) and Local Time fields now tick live once per second while idle; seconds are optional when typing a manual/backdated time by hand.
1.30	Restored the Comment field to the QSO Entry form; default station profile shown in bold; fixed a startup crash caused by a stale saved log location.
1.29	Entry form, Edit QSO, Settings, and Station Profiles windows now scale with window size; Settings redesigned (14pt, two columns); Radio Control section

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	clarified.
1.28	QSO Entry/Edit forms: manual UTC/Local Date-Time entry with sync, SOTA/POTA fields, reordered/larger entry form. Log grid column widths now persist. Major ADIF 3.1.4 compliance fixes: correct byte-length field encoding (fixed real name corruption from a QRZ export), correct QSL_SENT/QSL_RCVD enumerations, correct Band token for 1.25M, and Mode/Sub-Mode now match ADIF's real vocabulary directly (no more synthetic DATA mode).
1.27	Version bump for a fully automated, unattended publish.ps1 run (no app behavior changes). publish.ps1's manual-PDF export now runs with a 90-second timeout guard so a stuck Word export can no longer hang the whole publish.
1.26	Main window banner now reads "Conejo Valley Amateur Radio Club Logger" instead of "CVARC LOGGER". publish.ps1's output folder renamed to "CvarcLogger" (from "win-x64"), and the published zip now includes a PDF of this User Manual.
1.25	Multi-select QSOs in the log grid (Ctrl/Shift-right-click) with bulk delete; copy/paste QSO rows via Ctrl+C / Ctrl+V.
1.24	ADIF import reads QRZ Logbook's own confirmation fields; larger banner/title; "Program by W6CSM" credit added to the status bar.
1.23	Fixed a Station Profile double-save crash; ADIF import no longer aborts on unrecognized DXCC codes; merged Comment/Notes into one Comment field; input masks added to date/time fields; database now created next to the exe for portable installs.
1.20	Row numbers added to the log grid; Enter key now logs a QSO; column order is now draggable and remembered.
1.19	App icon and banner logo added.
1.18	OP field now prefers the station profile's Operator Callsign over its free-text OP name.
1.17	Added 1.2M band; SSB Sub-Mode with CAT auto-fill;

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	ARRL Section auto-derived from State/County; Station Profile UTC Offset/DST fields for Local Time; Edit QSO window made fully editable and resizable; QTH/OP fields added.

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Appendix B: Credits

CVARC Logger is developed and maintained by W6CSM.

Beta Testers

- Norman Campbell, AB6ET
- Greg Lane, K7SDW
- Keith Elliot, W6KME
- Jeff Reinhardt, AA6JR

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